

BASIC CHEMISTRY OF LIVING THINGS

Biology 9

Mr. Queenan

Text - Ch 2.1-2.2 (p 42-51)

Section 1: Composition of Matter

1: Matter

- Understanding chemistry is important to biologists:
 - ▣ Living things composed of the same kinds of matter as non-living things
 - ▣ Knowing how matter changes leads to understanding of life processes of organisms
- **Matter:** anything that occupies space & has mass
- **Mass:** quantity of matter in an object
- **Weight:** force produced by gravity acting on mass
 - ▣ Mass $\neq$ weight

1: Elements

- Substance that can not be converted into a simpler substance with different properties (density, solubility...)
- More than 100 elements identified (& continue to be identified)
 - ▣ ≈ 30 are involved in living processes
 - Primarily O, C, H, N
- Elements organized on the **periodic table**
- Each has own **chemical symbol**:
 - ▣ 1-3 letters; some are Latin abbr.
 - ▣ Atomic number
 - ▣ Atomic mass

Atomic number	→ 6	
	C	← Elemental symbol
Element name	→ Carbon	
	12.01	← Atomic weight

1: Atoms

- **Atom:** smallest component of an element that retains its properties
 - ▣ Atomic properties determine structure and properties of matter they compose
 - ▣ Cannot be directly observed; understood through models
- Atomic nucleus contains most of the mass of the atom; 2 types of sub-atomic particles: **proton** (+ charge) and **neutron** (0 charge)

1: Atoms

- **Atomic number:** number of protons in the atom
- **Atomic mass (Mass Number):** number of protons + number of neutrons in the atom
 - ▣ Together, these can be used to determine the number of neutrons in the nucleus
- Amount of charge in the atom is balanced
 - ▣ Number of protons (+) = number of electrons (-)
 - ▣ Atoms are neutral: 0 net charge

1: Atoms

- **Electrons:** high energy, negatively charged particles with negligible mass
 - Revolve around the nucleus in defined paths known as **orbitals**
 - 3D region around the nucleus indicating the probable location of an electron
 - Move so quickly – if possible to observe, would not look like objects, but an electron cloud (blur)
 - Electrons further from the nucleus have more energy than those close to the nucleus

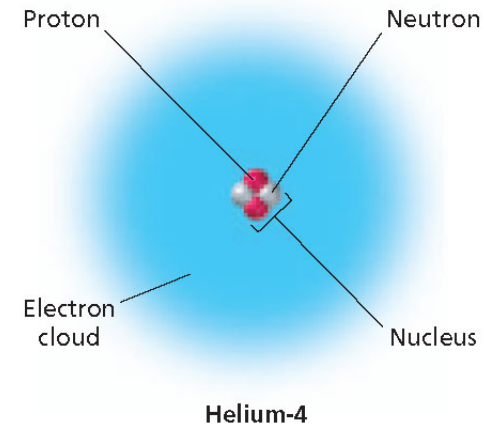
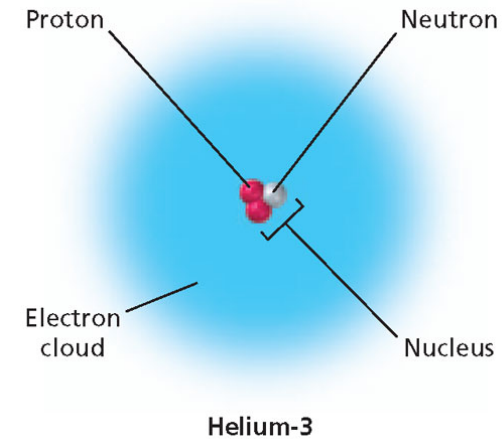
1: Atoms

Periodic Table of the Elements

1 IA H Hydrogen 1.008 1	2 IIA He Helium 4.003 2																	13 IIIA B Boron 10.81 5	14 IVA C Carbon 12.01 6	15 VA N Nitrogen 14.01 7	16 VIA O Oxygen 16.00 8	17 VIIA F Fluorine 18.99 9	18 VIIIA Ne Neon 20.18 10
3 Li Lithium 6.94 3	4 Be Beryllium 9.01 4																	19 Al Aluminum 26.98 13	20 Si Silicon 28.09 14	31 P Phosphorus 30.97 15	32 S Sulfur 32.06 16	33 Cl Chlorine 35.45 17	34 Ar Argon 39.95 18
5 Na Sodium 22.99 11	6 Mg Magnesium 24.31 12	3 IIIB Sc Scandium 44.96 21	4 IVB Ti Titanium 47.88 22	5 VB V Vanadium 50.94 23	6 VIB Cr Chromium 51.99 24	7 VIIB Mn Manganese 54.94 25	8 VIIIB Fe Iron 55.85 26	9 VIIIB Co Cobalt 58.93 27	10 VIIIB Ni Nickel 58.69 28	11 Cu Copper 63.55 29	12 IIB Zn Zinc 65.38 30	35 Ga Gallium 69.72 31	36 Ge Germanium 72.64 32	37 As Arsenic 74.92 33	38 Se Selenium 78.96 34	39 Br Bromine 79.90 35	40 Kr Krypton 83.80 36						
7 K Potassium 39.10 19	8 Ca Calcium 40.08 20	39 Y Yttrium 88.91 39	40 Zr Zirconium 91.22 40	41 Nb Niobium 92.91 41	42 Mo Molybdenum 95.94 42	43 Tc Technetium 98.91 43	44 Ru Ruthenium 101.07 44	45 Rh Rhodium 102.91 45	46 Pd Palladium 106.42 46	47 Ag Silver 107.87 47	48 Cd Cadmium 112.41 48	49 In Indium 114.82 49	50 Sn Tin 118.71 50	51 Sb Antimony 121.76 51	52 Te Tellurium 127.60 52	53 I Iodine 126.91 53	54 Xe Xenon 131.29 54						
9 Rb Rubidium 85.47 37	10 Sr Strontium 87.62 38	57-71 Lanthanides	72 Hf Hafnium 178.49 72	73 Ta Tantalum 180.95 73	74 W Tungsten 183.84 74	75 Re Rhenium 186.21 75	76 Os Osmium 190.23 76	77 Ir Iridium 192.22 77	78 Pt Platinum 195.08 78	79 Au Gold 196.97 79	80 Hg Mercury 200.59 80	81 Tl Thallium 204.38 81	82 Pb Lead 207.2 82	83 Bi Bismuth 208.98 83	84 Po Polonium 209 84	85 At Astatine 210 85	86 Rn Radon 222 86						
11 Cs Cesium 132.91 55	12 Ba Barium 137.33 56	72 Rf Rutherfordium 261 104	105 Db Dubnium 262 105	106 Sg Seaborgium 266 106	107 Bh Bohrium 264 107	108 Hs Hassium 277 108	109 Mt Meitnerium 268 109	110 Ds Darmstadtium 271 110	111 Rg Roentgenium 272 111	112 Cn Copernicium 285 112	113 Nh Nihonium 284 113	114 Fl Flerovium 289 114	115 Mc Moscovium 288 115	116 Lv Livermorium 293 116	117 Ts Tennessine 289 117	118 Og Oganesson 294 118							
13 Fr Francium 223 87	14 Ra Radium 226 88	89-103 Actinides	104 Rf Rutherfordium 261 104	105 Db Dubnium 262 105	106 Sg Seaborgium 266 106	107 Bh Bohrium 264 107	108 Hs Hassium 277 108	109 Mt Meitnerium 268 109	110 Ds Darmstadtium 271 110	111 Rg Roentgenium 272 111	112 Cn Copernicium 285 112	113 Nh Nihonium 284 113	114 Fl Flerovium 289 114	115 Mc Moscovium 288 115	116 Lv Livermorium 293 116	117 Ts Tennessine 289 117	118 Og Oganesson 294 118						
57 La Lanthanum 138.91 57	58 Ce Cerium 140.12 58	59 Pr Praseodymium 140.91 59	60 Nd Neodymium 144.24 60	61 Pm Promethium 145 61	62 Sm Samarium 150.36 62	63 Eu Europium 151.96 63	64 Gd Gadolinium 157.25 64	65 Tb Terbium 158.93 65	66 Dy Dysprosium 162.50 66	67 Ho Holmium 164.93 67	68 Er Erbium 167.26 68	69 Tm Thulium 168.93 69	70 Yb Ytterbium 173.05 70	71 Lu Lutetium 174.96 71									
89 Ac Actinium 227 89	90 Th Thorium 232.04 90	91 Pa Protactinium 231.04 91	92 U Uranium 238.03 92	93 Np Neptunium 237.05 93	94 Pu Plutonium 244.06 94	95 Am Americium 243.06 95	96 Cm Curium 247 96	97 Bk Berkelium 247.07 97	98 Cf Californium 251 98	99 Es Einsteinium 252 99	100 Fm Fermium 257 100	101 Md Mendelevium 258 101	102 No Nobelium 259 102	103 Lr Lawrencium 262 103									

1: Atoms & Isotopes

- All atoms of the same element have the same number of protons (constant)
 - Number of neutrons can vary
- **Isotope:** variants of element with differing number of neutrons
 - Change the mass of the element
 - Most elements composed of a mixture of isotopes
 - Atomic mass found in periodic table is actually average atomic mass which accounts for relative amounts of each isotope



1: Compounds

- Elements not usually found alone
- Atoms readily combine with other atoms of the same or different elements to form **compounds**
 - ▣ Made of atoms of 2 or more elements in fixed proportions
- **Chemical formula** shows type and proportion of atoms in a compound
 - ▣ H_2O , CO_2 , H_2 , CH_4 , CH_3OH
- Physical and chemical properties of compounds and their component elements are different
 - ▣ $\text{O}_2 (\text{g}) + 2\text{H}_2 (\text{g}) \rightarrow 2\text{H}_2\text{O} (\text{l})$

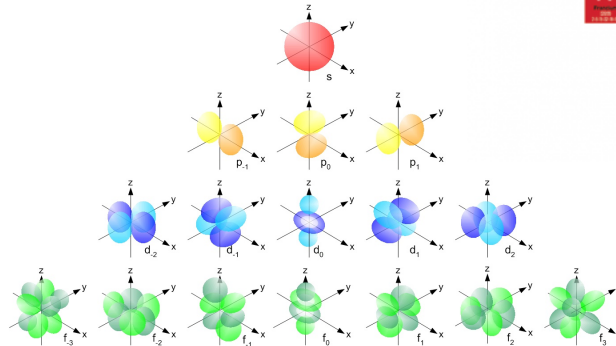
1: Compounds

- How elements combine to form compounds depends on number and arrangement of their electrons
- Atoms are most stable when highest energy orbital is full with electrons
 - ▣ Known as **valence orbital**
- Elements, such as He & Ne, that already have the maximum number of electrons in their valence orbitals are known as **noble/inert**; don't react with other elements
 - ▣ Most elements not in stable form; react to add stability

Nucleus

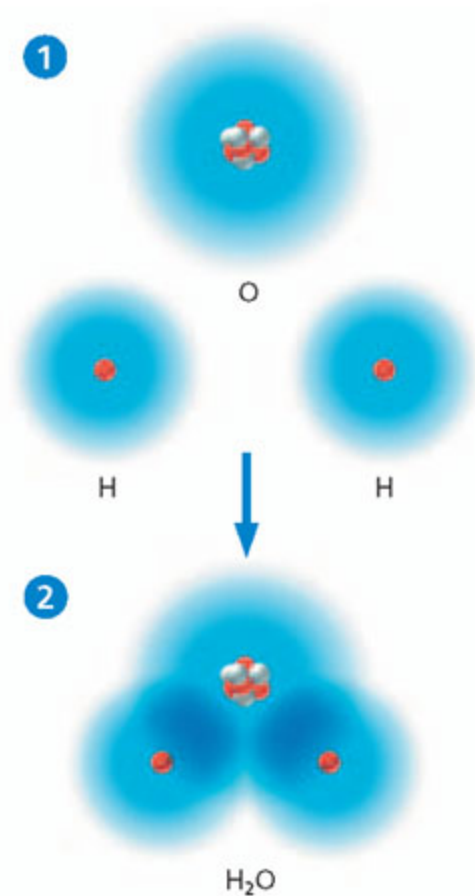


1 IA	Atomic Number → 1 H ← Spinless Name → Hydrogen Electrons per shell → 1 Atomic Weight																18 VIIIA	He										
2 IIA	Li Lithium 6.941	Be Beryllium 9.012															13 IIIB	B Boron 10.811	14 IIIVA	C Carbon 12.011	15 VA	N Nitrogen 14.007	16 VIA	O Oxygen 15.999	17 VIIA	F Fluorine 18.998	18 VIIIA	Ne Neon 20.180
State of matter (color of name)			Subcategory in the metal-metalloid-semimetalloid (color of background)			Legend			Unlabeled chemical properties																			
(S) SOLID (L) LIQUID (G) GASEOUS			(M) METAL (ML) METALLOID (S) SEMIMETALLOID			(L) LANTHANIDES (S) SOLUBLE IN WATER (N) NONMETALS																						



1: Chemical Bonds - Covalent

- When elements react/interact, they form a **chemical bond**
 - ▣ Attractive force that hold atoms together
- Major forms of chemical bonds: **covalent & ionic**
- **Covalent bond:** when two atoms share one or more pairs of electrons
 - ▣ Ex: H_2O – each H needs 1 electron to become stable; O needs 2 electrons to become stable; all become stable by sharing electrons
- Most covalent bonds between 2 non-metals
- Lead to formation of **molecules**: smallest amount of a substance that retains its properties and can exist in a free state



1: Chemical Bonds - Ionic

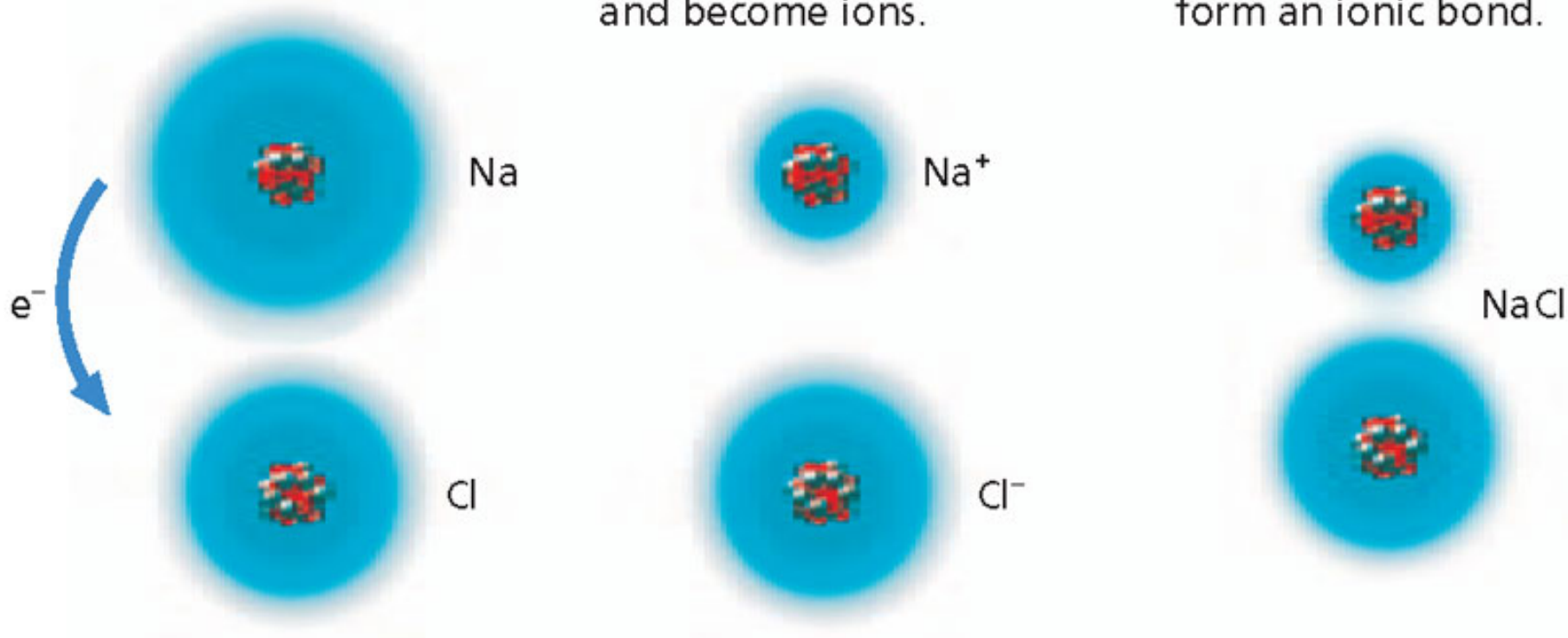
- **Ionic bond:** one atom transfers an outer electron to another atom, resulting in 2 ions
 - Ion: an atom or molecule with an electrical charge (+ or -)
 - Example ?
 - $\text{Na} \rightarrow \text{Na}^+ + \text{e}^-$; $\text{Cl} + \text{e}^- \rightarrow \text{Cl}^-$
- Positive and negative charges attract – this attractive force known as **ionic bond**
- Most ionic bonds are between metals and non-metals and result in “salts”

2-1: Chemical Bonds - Ionic

1 A Na atom loses an electron to a Cl atom.

2 Both atoms are now stable, but carry a charge and become ions.

3 The ions are attracted to one another to form an ionic bond.



1: Chemical Bonds

- Although these are the two most common, they are not the only types of bonds.
- We will discuss additional chemical bonds when we discuss water and all of its properties

Section 2: Energy

2: Energy & States of Matter

- **Energy:** ability to do work
 - ▣ Cannot be created or destroyed, only converted between different forms
 - Ex: Electrical, radiant, thermal, kinetic, potential, chemical, mechanical
- **States of matter:** all matter is in constant motion that cannot be observed
 - ▣ Amount of motion and spacing between atoms determine its state:
 - **Solid** – closest in space, least amount of movement, fixed volume & shape
 - **Liquid** – medium amount of space, medium amount of movement, fixed volume, variable shape
 - **Gas** – most amount of space, most movement, variable shape & volume
- **Thermal energy** must be added to a substance to change its state

2: Chemical Reactions

- **Chemical reaction:** one or more substances change to produce one or more different substances
 - Energy is absorbed or released when chemical bonds are broken and new ones are formed
 - Generally written with starting material (**reactants**) on the left and the new material (**products**) on the right
 - When the arrow is uni-directional ($\rightarrow$) reaction only occurs in one direction (irreversible)
 - When arrow is bi-directional ($\leftrightarrow$) reaction can occur in either direction (reversible)
- Occur constantly in the body: energy supplying substances are broken down to CO_2 , H_2O , other products; releasing energy for cells to do work
 - Sum = metabolism

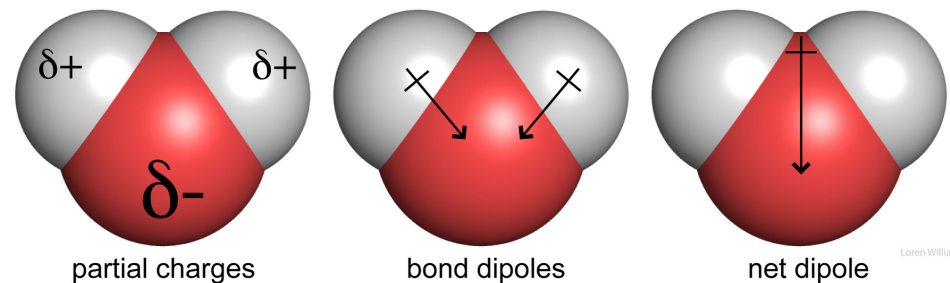
2: Change

- There are two main types of change that matter can undergo:
 - Physical Change – Matter changes form but not chemical identity
 - Water Boiling, freezing, ice being chopped, metal being bent
 - Chemical Change – Matter undergoes a chemical reaction
 - Wood burning, food being cooked, iron rusting

Section 3: Water & Solutions

3: Polarity of Water

- Biological function of water comes from its chemical structure
 - H's and O share electrons in covalent bonds – **unequally**
 - O attracts more electrons than H, pulling them toward its own nucleus
- Although molecule has net 0 charge, each atom has partial charge
 - O is partially $-$ (δ^-) and Hs are partially $+$ (δ^+). When charge is not equally distributed among atoms of a molecule, known as **polar**



3: Solutions & Solubility

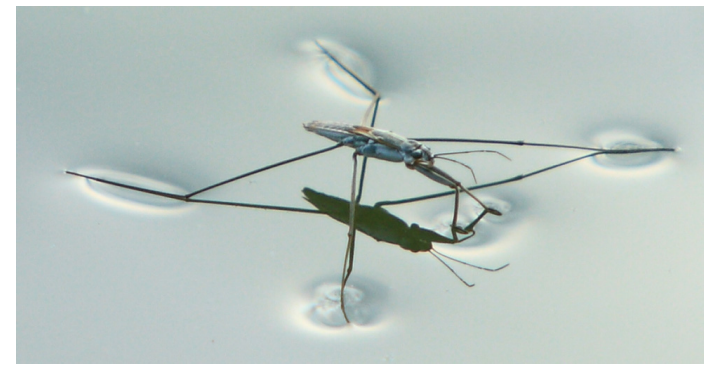
- **Solubility:** ability of a substance to dissolve another substance
 - **“Like dissolves like”:** polar substances dissolve other polar substances (sugars, ionic compounds, some proteins); does not dissolve non-polar substances (oil, other substances with high CH content)
 - Ex: NaCl (ionic compound) can dissolve in water
 - Na^+ attracted to $\sigma^- \text{O}$; Cl^- attracted to $\sigma^+ \text{H}$
- Importance?
 - Ions (esp. Na^+ , Cl^- , K^+ , Ca^{2+}) necessary for essential body functions: **muscle contraction, transmission of impulses** throughout nervous system

3: Hydrogen Bonding

- Polarity of water causes its molecules to be attracted to one another
 - ▣ $\delta^- \text{O}$ attracted to $\delta^+ \text{H}$ of another molecule
 - ▣ This attractive force known as a **hydrogen bond**
- H bonding causes water to “cling” to itself and some other substances
 - ▣ These bonds constantly form, break, re-form
 - ▣ H_2O (s) – all molecules are H bonded and do not break
 - ▣ H_2O (l) – addition of energy breaks some H bonds until number of **bonded and unbonded molecules are equal**
 - ▣ H_2O (g) – addition of energy breaks remainder of H bonds until **few to none remain**
- H bonding responsible for **special properties of water:**
 - ▣ Cohesion, adhesion, heat capacity, evaporative cooling, density of ice, solubility

3: Cohesion

- **Cohesion:** attractive force that holds molecules of a single substance together
 - ▣ Ability of substance to stick to itself
 - ▣ Due to H bonding – allows upward movement of water from plant roots to leave
- Related to cohesion – **surface tension**
 - ▣ Cohesive forces resulting from H bonds cause molecules at the surface of water to be pulled downward into the liquid
 - ▣ Result – water acts as if it has thin “skin” over the surface
 - Ex: in an over-filled glass, water appears to bulge over the rim
 - Ex: small creatures are able to run on water’s surface without sinking



3: Adhesion

- **Adhesion:** attractive force between particles of different substances
 - Ex: water and glass
- Related to adhesion – **capillarity**
 - Attraction between molecules resulting in rise of a surface of liquid when in contact with a solid
- Cohesion, adhesion, capillarity work together to help water rise through narrow tubes against the force of gravity
 - Ex: plant vascular system

3: Heat Capacity

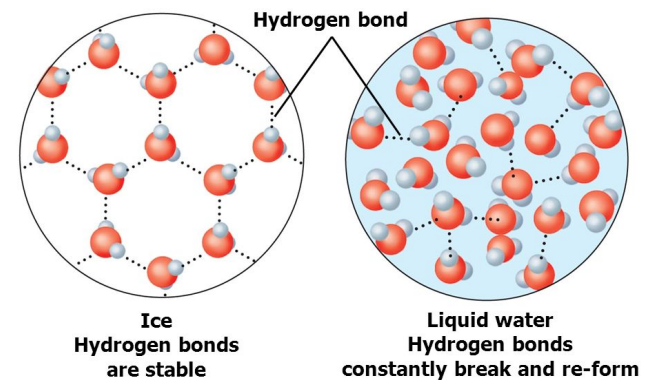
- Water has **high heat capacity**
 - ▣ Ability to **absorb or release** relatively **large amounts of heat** energy with only a **slight change in temperature**
 - ▣ Due to H bonding – energy is **absorbed to break H bonds** and is **released when H bonds form**
- First, energy input **breaks H bonds**, then **increases movement of molecules** causing **increase in temperature**
 - ▣ When temperature drops, H bonds re-form and release heat energy
 - Ex: Heat from sun absorbed by water which cools the air; at night, gradual cooling of water warms the air

3: Evaporative Cooling

- Heat capacity of water:
 - Allows oceans to stabilize temperature of Earth such that life can survive
 - Allows cells to maintain temperature despite changes in environmental temperature
- **Evaporative cooling:** relatively large amount of energy absorbed by water during evaporation which significantly cools surface of remaining liquid
 - Prevents organisms from over-heating
 - Ex: sweating, panting

3: Density of Ice

- **Density:** amount of matter per unit area
- **Solids:** most dense of all forms of matter; except for solid water
 - ▣ Ice is LESS dense than liquid water due to the shape of water molecule and H bonding
 - **Wide angle between H atoms** allows formation of ice crystals with large amounts of open space (i.e. low density)
- Allows ice to float in liquid water
 - ▣ Ponds and lakes freeze top → bottom; insulating water below and allowing aquatic organisms to survive



3: Solutions

- **Solution:** mixture in which one or more substances are **uniformly distributed** in another substance
 - ▣ Can be a mixture of solid, liquid or gas
- Parts of a solution:
 - ▣ **Solute:** a substance dissolved IN the solvent (the substance that is, itself, dissolved)
 - Can be ions, atoms or molecules
 - ▣ **Solvent:** substance IN WHICH the solute is dissolved (the substance that does the dissolving)
 - Ex: sugar water – sugar = solute; water = solvent
 - ▣ Molecules can be **hydrophobic** or **hydrophilic**
 - Hydro – water; phobic – afraid of; philic - loves

3: Aqueous Solutions

- **Aqueous solution:** water is the solvent
 - ▣ Important to all living things
 - ▣ Ocean is a aqueous solution – marine life habitat
 - ▣ Nutrients for plants are in aqueous solution in soil
 - ▣ Cells exist in interstitial (intercellular) fluid, which is an aqueous solution
 - ▣ Cells filled with fluid that in an aqueous solution
- Most chemical reaction in the body occur in aqueous solution

3: Acids & Bases

- **Acidity & alkalinity** need to be balanced in living systems
- Acids & bases start with the **ionization of water**
 - Water molecules can collide with one another; sometimes the interaction is strong enough to cause a chemical change
 - Water molecule dissociates: $\text{H}_2\text{O} \leftrightarrow \text{H}^+ + \text{OH}^-$
 - H^+ = proton; OH^- = hydroxide ion
- **Acidity**: measure of relative amount of protons
- **Alkalinity**: measure of relative amount of hydroxide ion

3: Acids & Bases

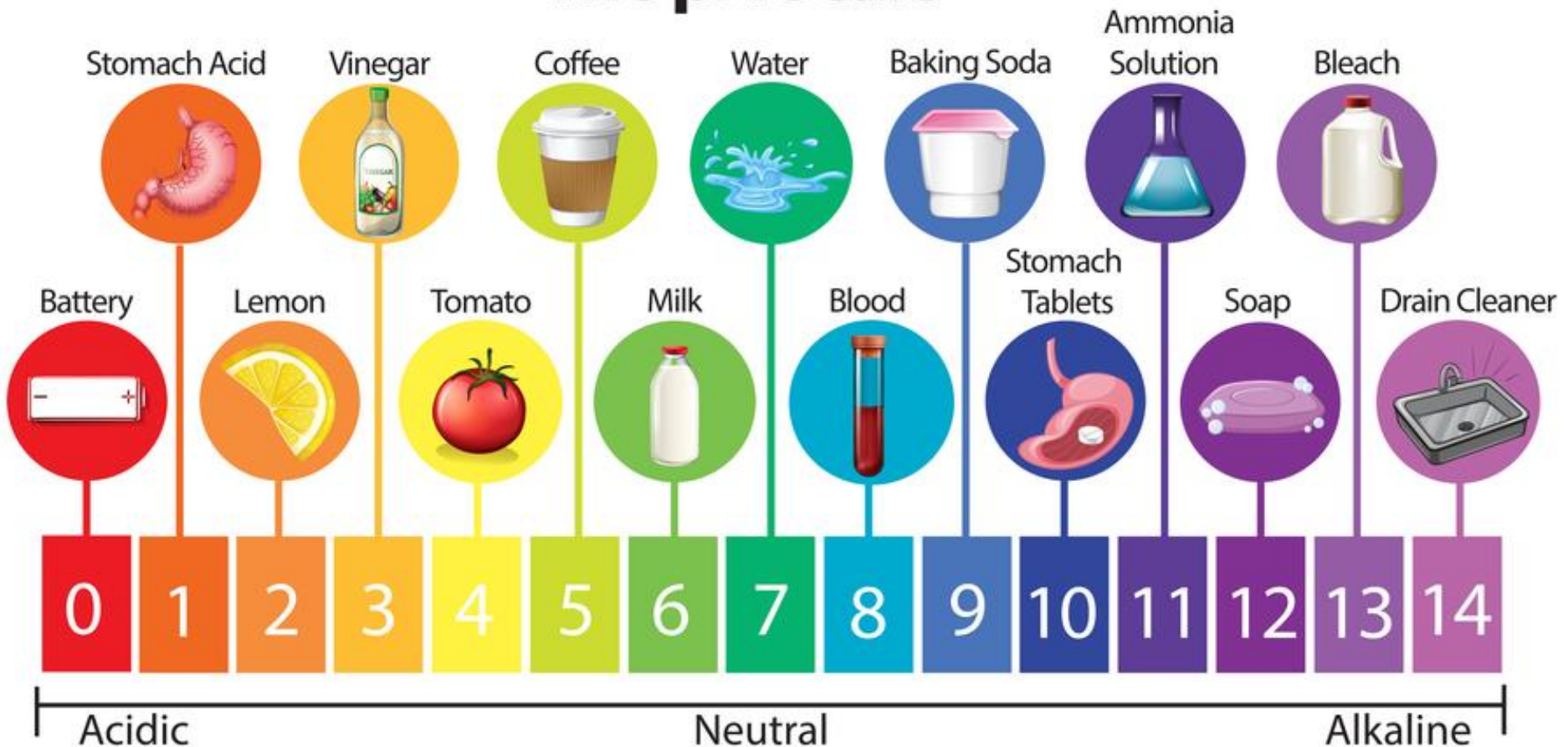
- When the amount of H^+ = **amount of OH^-** - solution is **neutral**; pure water
- When the amount of H^+ > **amount of OH^-** - solution is an **acid**
 - Ex: $\text{HCl (aq)} \leftrightarrow \text{H}^+ + \text{Cl}^-$
 - Leads to higher hydronium than hydroxide in solution
 - Acids tend to taste **sour**: vinegar (acetic acid); citrus fruit (citric acid)
- When the amount of OH^- > **amount of H^+** - solution is a **base**
 - Ex: $\text{NaOH (aq)} \leftrightarrow \text{Na}^+ + \text{OH}^-$
 - Leads to higher hydroxide than hydronium in solution
 - Bases described as “alkaline”; tend to be **bitter**
 - Feel slippery because OH^- reacts with oil in skin

3: pH

- pH scale used to compare **relative concentrations** of **protons and hydroxide** in solution
- Ranges from 0 – 14
 - ▣ 0 = very acidic; 7 = neutral; 14 = very alkaline
- pH scale is logarithmic; each unit is equal to a **10-fold difference**, not a 1-fold difference
- Measured by:
 - ▣ Litmus paper – changes color based on pH
 - ▣ Chemical indicator – changes color based on pH
 - ▣ pH meter – directly measures pH and displays on screen

3: pH

The pH Scale



3: Control of pH

- pH needs to be **tightly controlled** in living organisms
 - Enzymes & other proteins function only within **narrow pH range**
- Control of pH in living system accomplished by **buffers**
 - Chemical substance that can **neutralize** small amounts of acid or base added to the solution
- In blood, most important buffer system is **bicarbonate/carbonic acid** system
 - $\text{CO}_2 + \text{H}_2\text{O} \leftrightarrow \text{H}_2\text{CO}_3 \leftrightarrow \text{HCO}_3^- + \text{H}^+$